



Mahmoud EL-Khouly

Student Edition

Internet Technology

From Networks to Streaming Servers

LAP
LAMBERT
Academic Publishing

Internet Technology

Prepared By: Dr. Alaa Khalifa

Course Essentials: Code, Credits, and Grades

Course Identification

Course Code: IT223
Credit Hours: 3



Grading Distribution

Midterm	20
Assignments	10
Project	20
Final	50

Lecture Guidelines



No Side Talk



Active Participation



Silent Your Device



Internet Technology

Chapter 1: Internet and The World Wide Web

Prepared By: Dr. Alaa Khalifa

Agenda



Evolution of the Internet



Data Transmission Methods



Layered System: Internet, Intranet, and Extranet



Internet Access Technologies

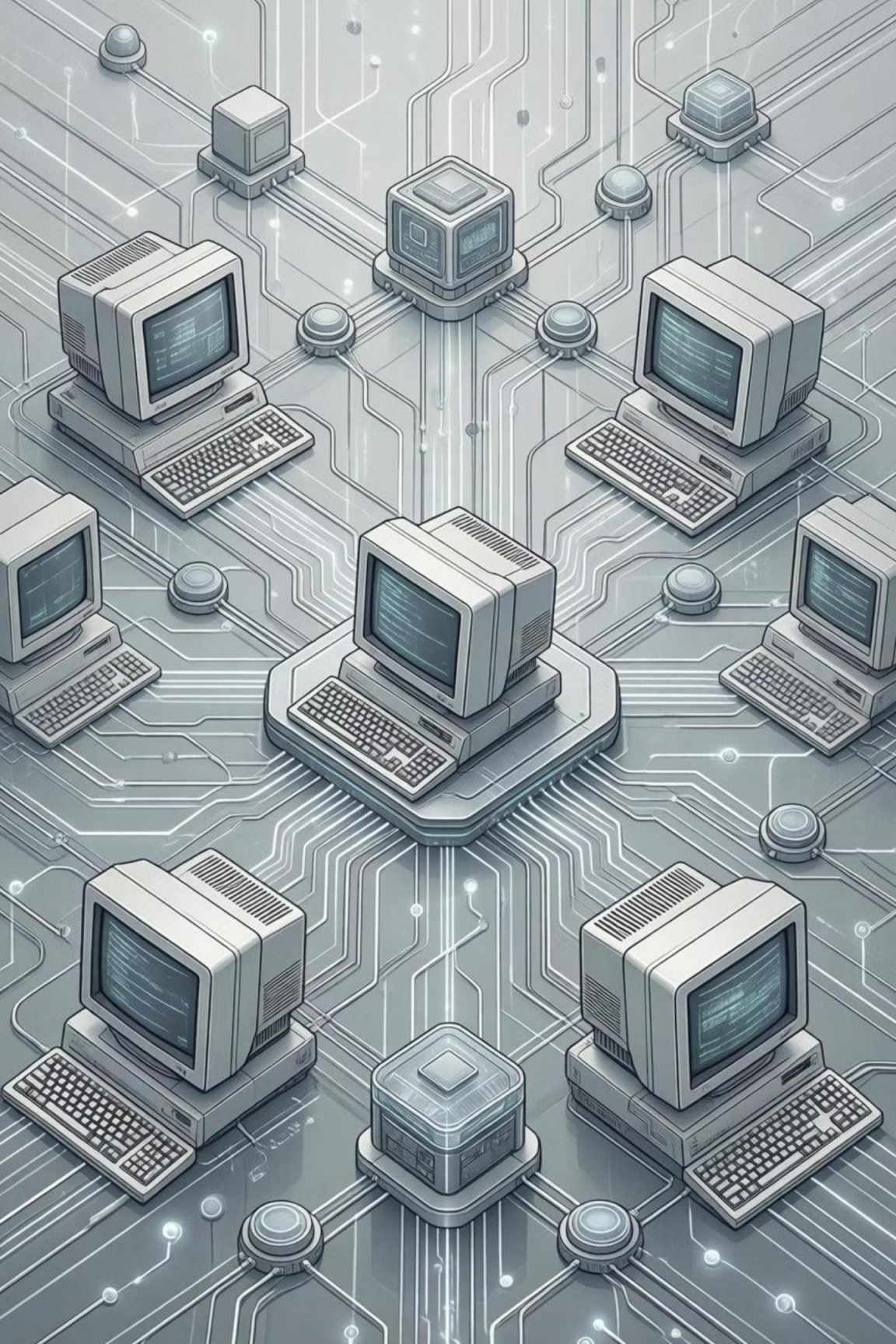


World Wide Web (WWW)



Internet Standards and Organizations



A detailed illustration of a network of vintage computers from the late 1960s or early 1970s. The computers, featuring large CRT monitors and keyboard units, are interconnected by a complex web of glowing blue circuit lines and nodes. The background is a light gray with a subtle grid pattern, and the overall aesthetic is futuristic and technical.

Evolution of the Internet

1969: ARPANET - The Birth of the Internet

Key Points

- Created by U.S. Department of Defense
- Connected 4 universities initially
- Introduced Packet Switching technology
- Designed for resilient communication

What is Packet Switching?

Breaking data into small packets that travel independently and reassemble at destination.

Impact

Foundation for modern internet infrastructure.

Evolution of the Internet

1970s–1980s: TCP/IP – The Universal Language

- Unified different networks into one interconnected system
- Enabled global internet expansion and standardization
- Two fundamental protocols that enable internet communication

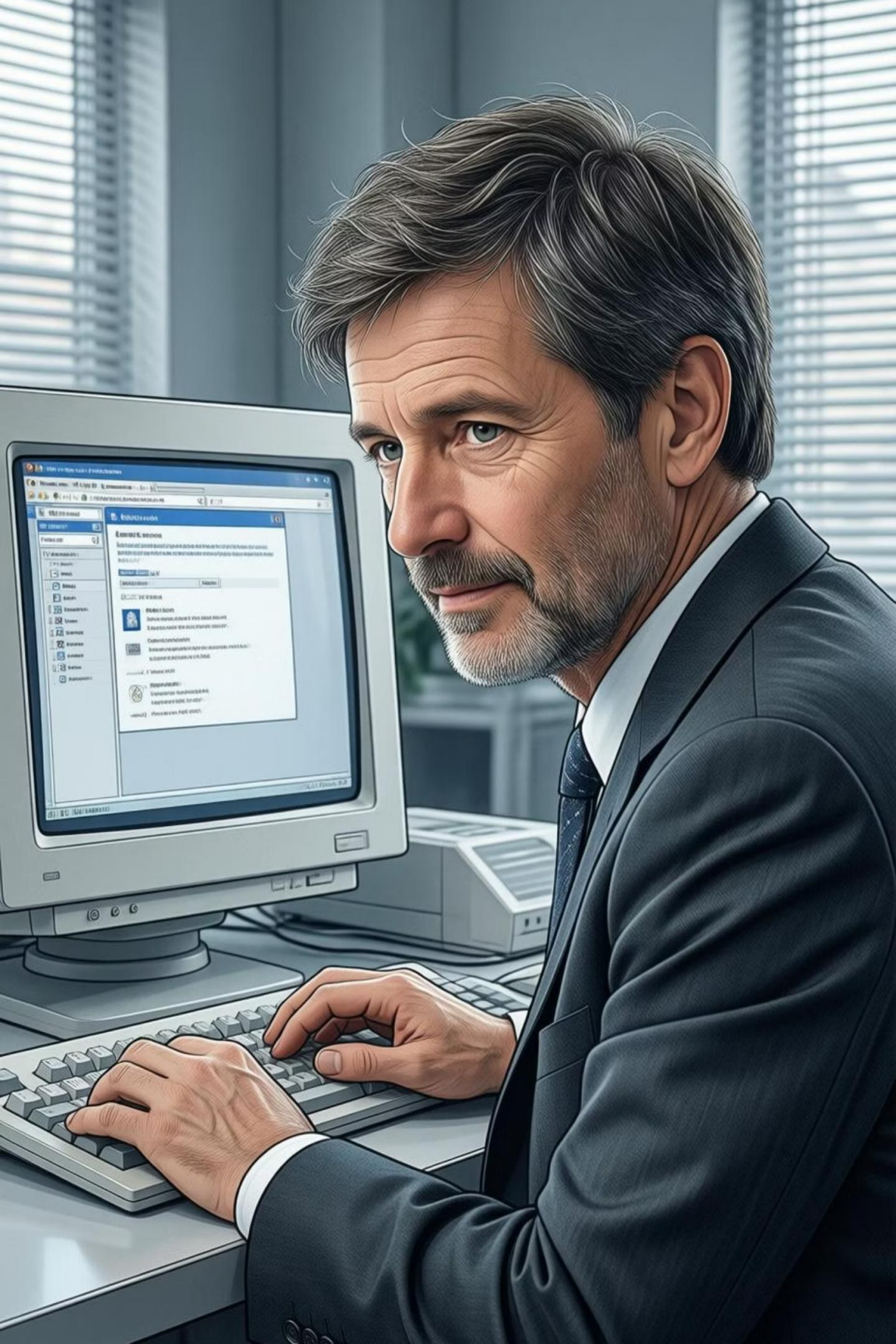
TCP (Transmission Control Protocol):

- Ensures reliable data delivery
- Checks that all packets arrive correctly
- Reassembles packets in proper order

IP (Internet Protocol):

- Routes packets to correct destination
- Assigns unique addresses to devices
- Handles packet addressing





Evolution of the Internet

1990s: Dot-Com & World Wide Web – Making the Internet Accessible

- Transformed the internet from academic/ military tool to global information platform
- Thousands of new ".com" companies launched
- Hypertext system for linking documents
- URLs (web addresses)
- HTML (markup language)
- Web browsers for easy access (Netscape Navigator, Internet Explorer)
- Made the internet user-friendly and accessible to non-technical people

Evolution of the Internet

2000s: Mobile Revolution

- 3G networks enabled mobile data
- 4G networks made it faster and more reliable
- Mobile internet became faster than desktop
- Internet became everywhere
- Created new industries (apps, mobile services)
- Social media exploded (Facebook, Twitter, Instagram)



Evolution of the Internet

2010s-Present: The Intelligent Internet – AI, Cloud & IoT



Cloud Computing

- Services like AWS, Azure, Google Cloud
- Store data and run applications online
- Accessible from anywhere, anytime



Internet of Things (IoT)

- Billions of connected devices
- Smart homes, smart cities
- Wearables, sensors, smart appliances



Artificial Intelligence & Machine Learning

- Powers recommendation systems (Netflix, YouTube)
- Virtual assistants (Siri, Alexa, Google Assistant)
- Predictive analytics and automation



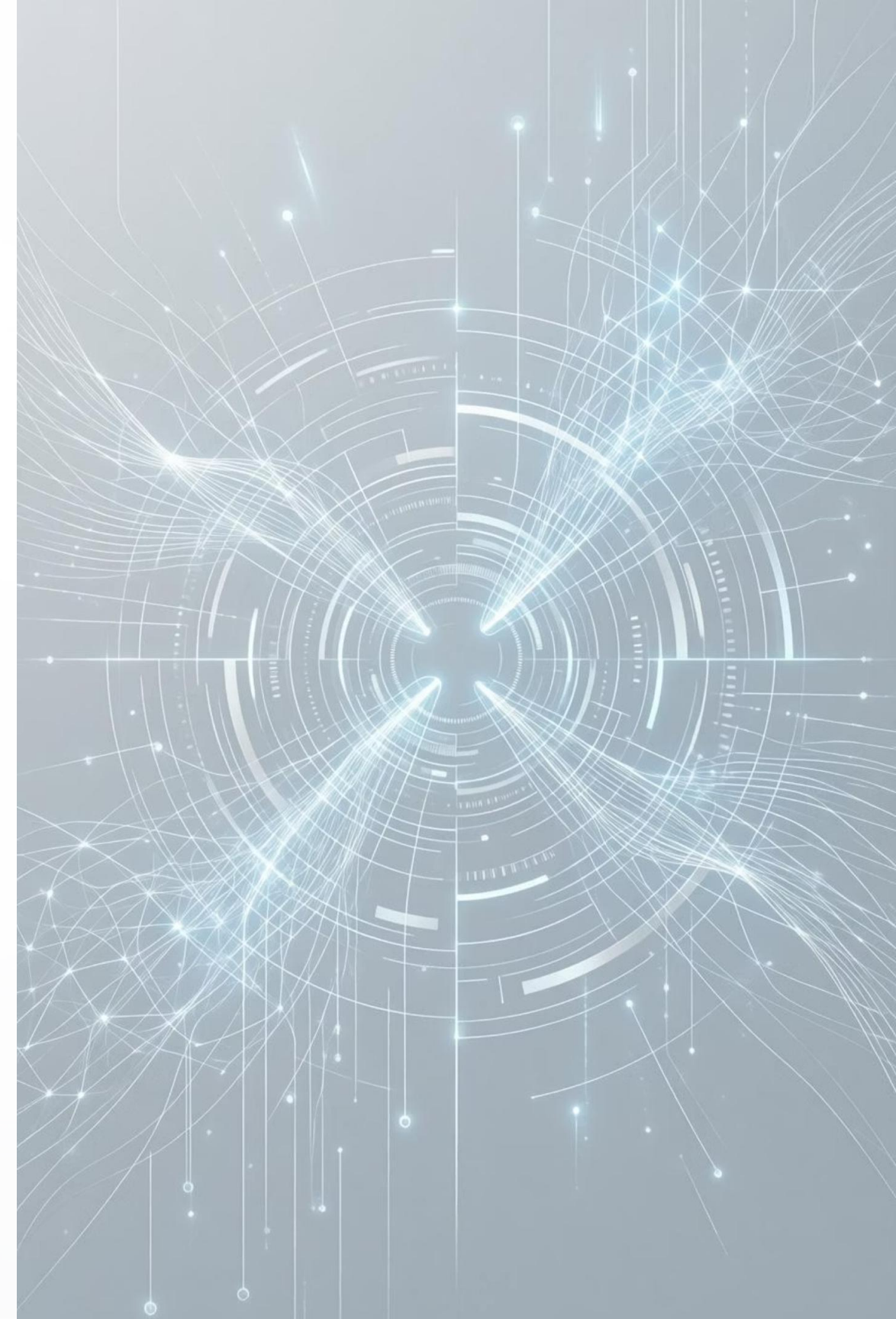
Big Data

- Massive amounts of data collected and analyzed
- Drives business decisions
- Powers personalization



Current Trends

- 5G networks for ultra-fast connectivity
- Edge computing for faster processing
- Cybersecurity becoming critical



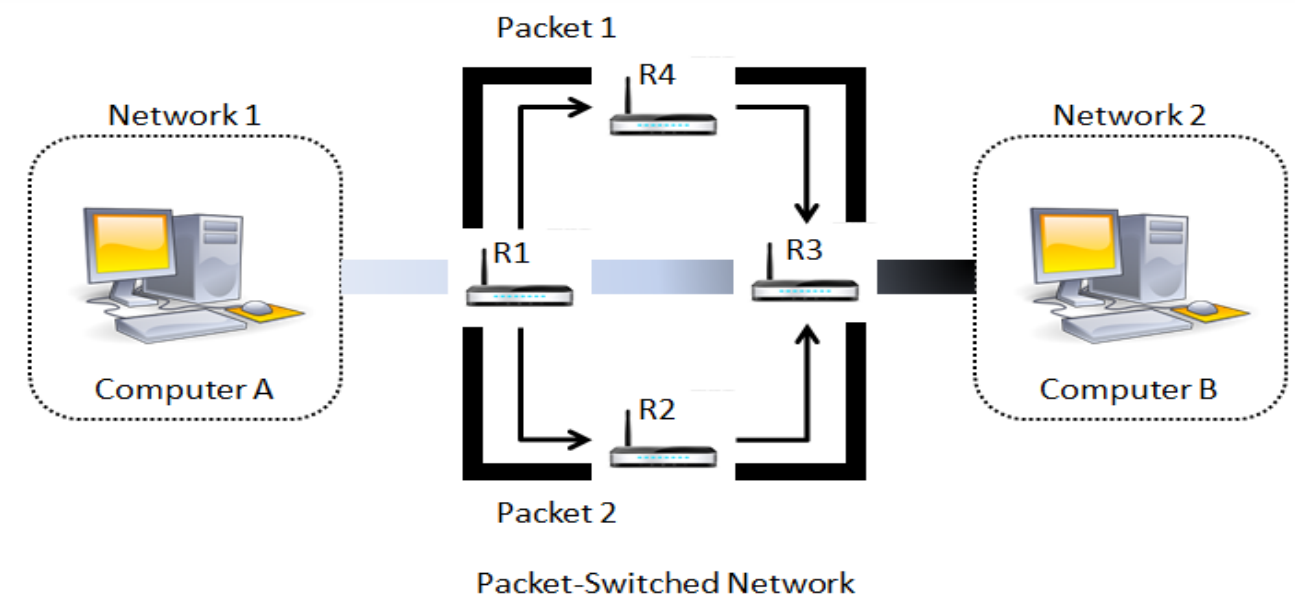
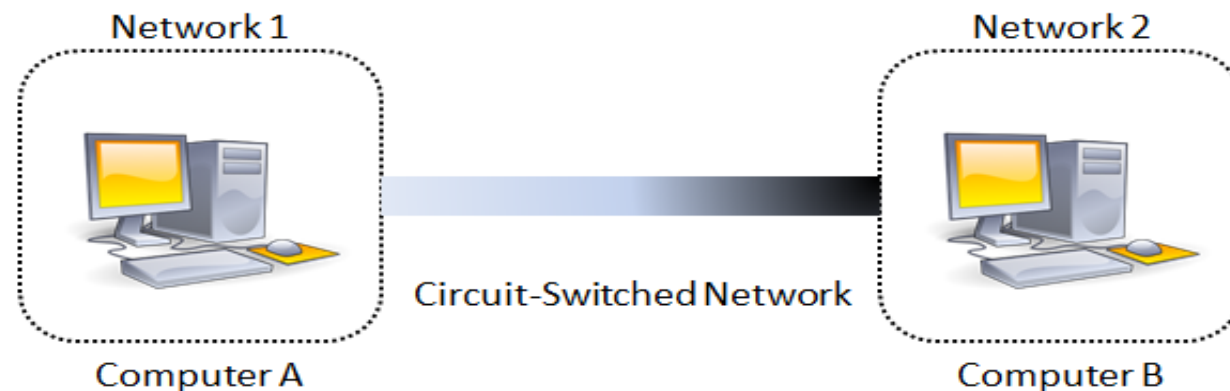
Data Transmission Methods

Data travels across networks using different transmission methods. Understanding these methods is crucial to comprehending how the Internet works.

Two Main Approaches:

Circuit Switching

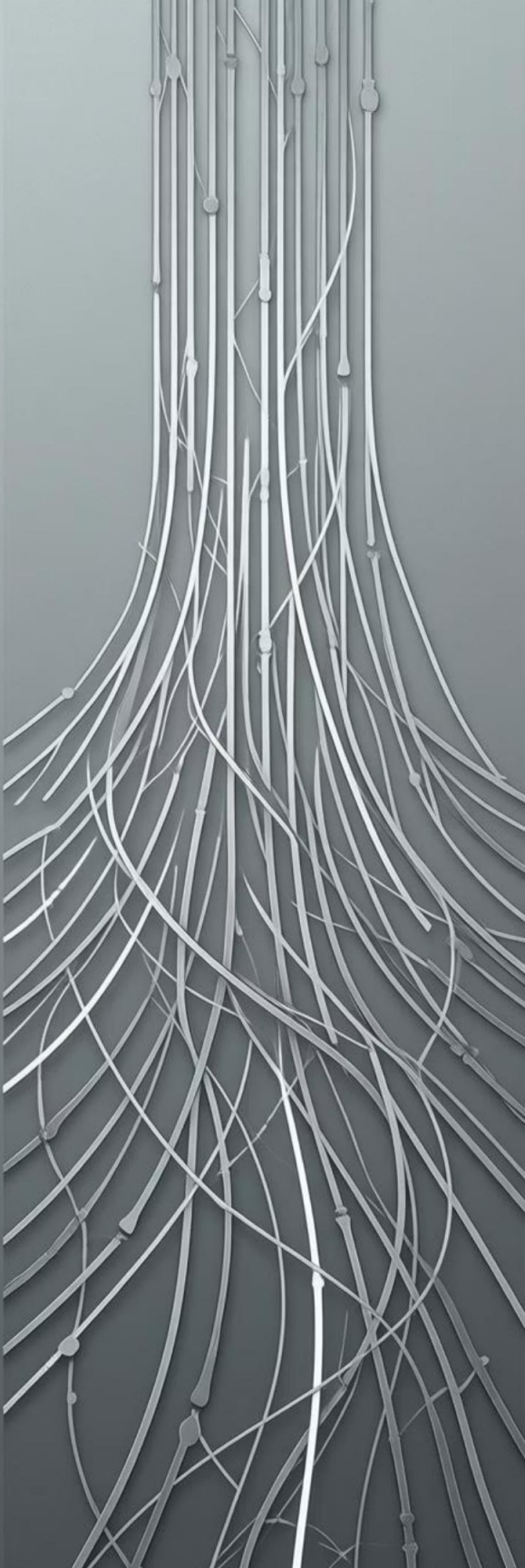
- In the case of a circuit-switched network : a direct physical connection is established between the sender and receiver.
- The data needs to be sent only through the connection that has been established. The other computers cannot use the communication channel until it is released.
- In the case of packet-switched networks : data that is to be transmitted, such as a file, is divided into small units of data called packets or data packets. The data packets can take different paths to reach the destination.
- An intermediate devices called routers can be used to identify the shortest path to a destination and transmit the data packets through that path.

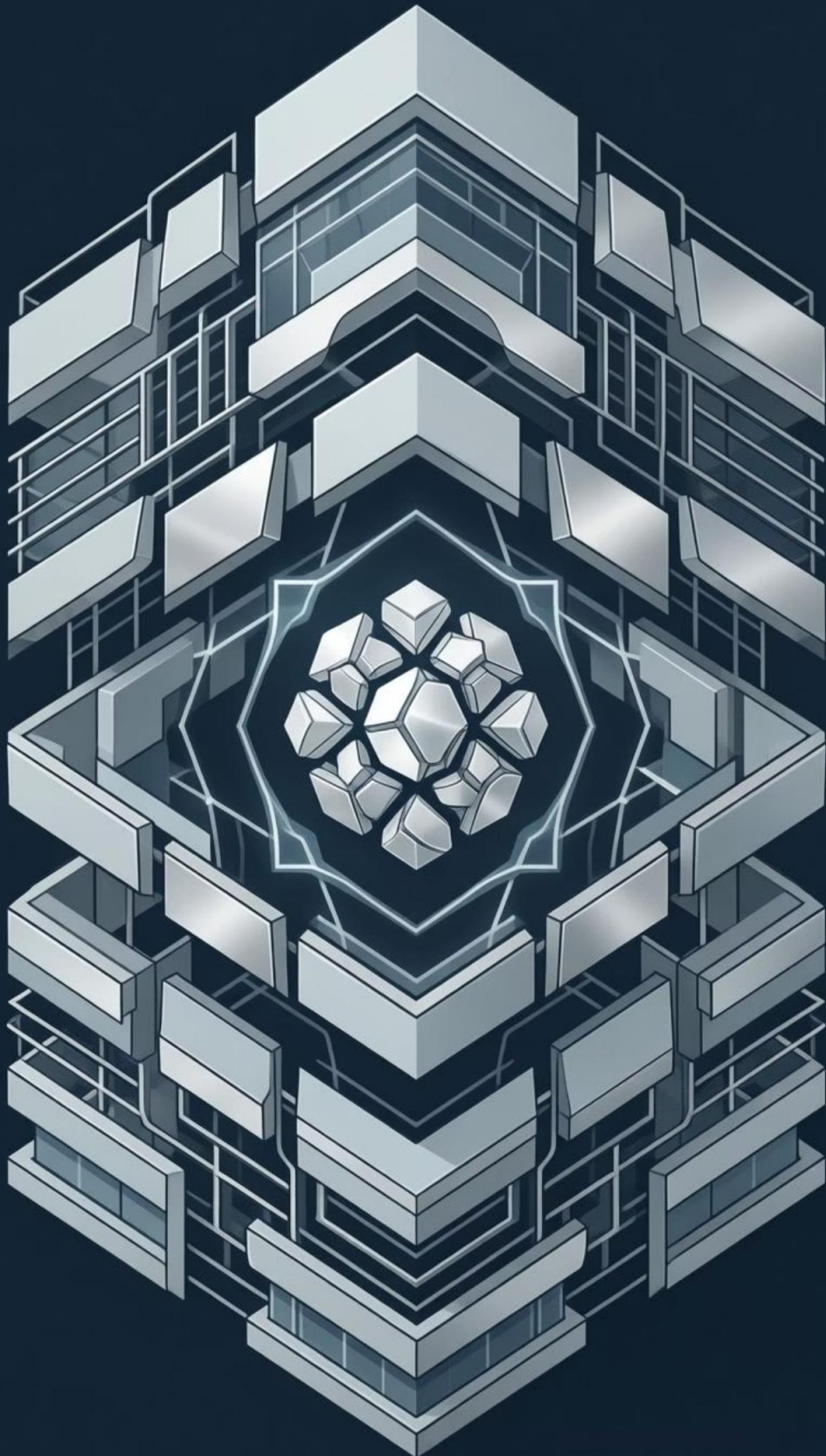


Data Transmission Methods

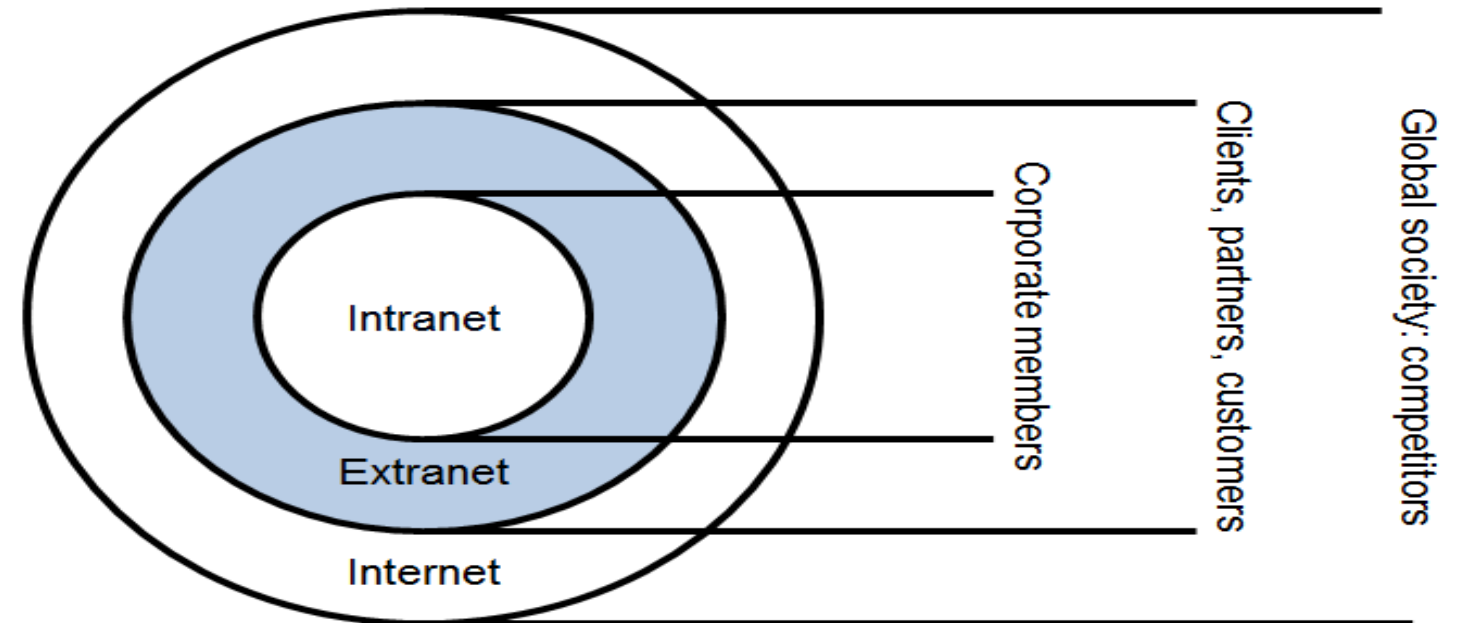
Comparison: Circuit Switching vs Packet Switching

Definition	Dedicated path	travel independently
Connection Setup	connection establishment	No connection setup needed
Path	Single dedicated path	Multiple possible paths
Bandwidth	Fixed and guaranteed	Dynamic and shared
Data Flow	Continuous stream	Discrete packets
Latency	Low and consistent	Variable
Efficiency	Lower (resources reserved)	Higher (resources used as needed)
Scalability	Limited	Excellent
Reliability	Depends on single path	Resilient to failures
Cost	Higher for long distances	Lower and more economical
Use Cases	Telephone calls, video conferencing	Internet, email, web browsing
Network Type	Traditional telephone networks	Modern Internet
Packet Loss	Rare	Possible under congestion





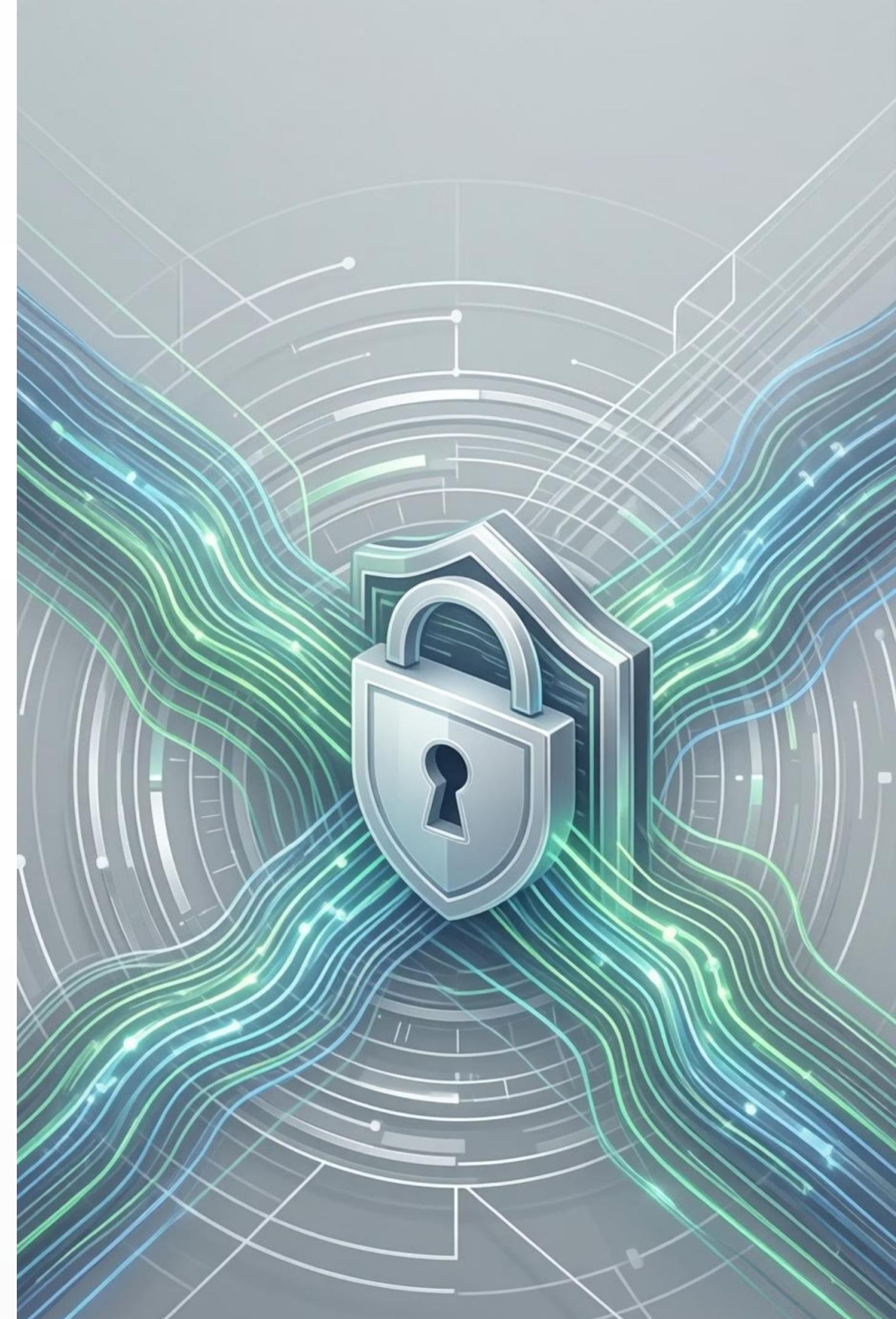
Layered System: Internet, Intranet, and Extranet



- Internet is the foundation that all others use
- Extranet extends Intranet to trusted external partners
- Intranet is built on Internet technologies but is private
- Each layer adds more security and restrictions
- Intranet is most secure, Internet is least secure

Layered System: Internet, Intranet, and Extranet (Intranet)

- An Intranet is a private network within an organization that uses Internet technologies but is restricted to authorized users only.
- It Uses same TCP/IP protocols as the Internet
- Operates behind firewalls for protection
- Requires login credentials for access
- Connects internal computers and servers
- **Key Characteristics:**
 - **Scope:** Organization-wide (single company or institution)
 - **Access:** Restricted - only employees and authorized users
 - **Users:** Company employees only
 - **Security:** High security with firewalls and authentication
 - **Ownership:** Owned and maintained by the organization



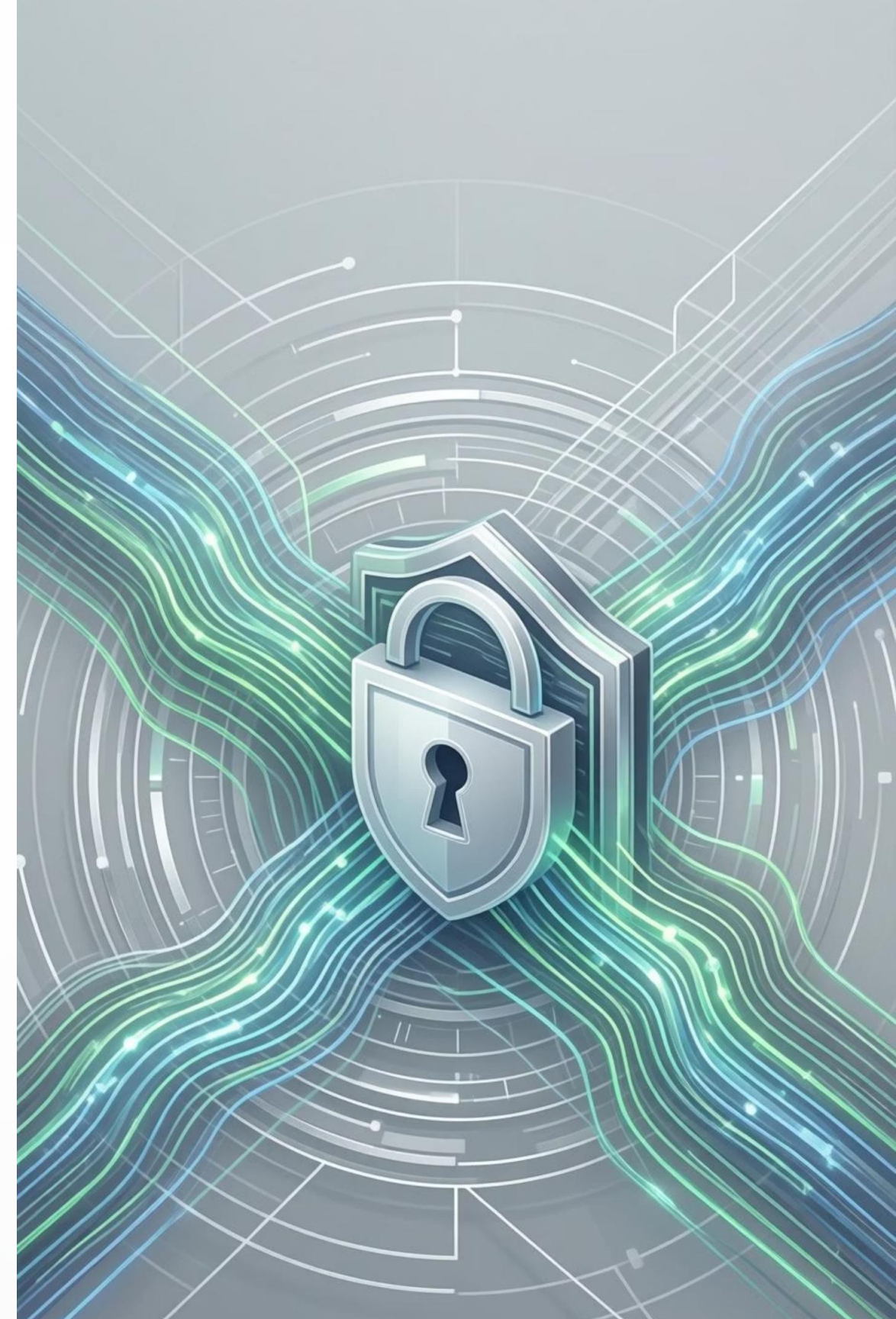
Layered System: Internet, Intranet, and Extranet (Intranet)

- **Common Uses:**

- Internal email and messaging
- Company policies and procedures
- HR systems and payroll
- Project management tools
- Document sharing and collaboration
- Internal announcements and news

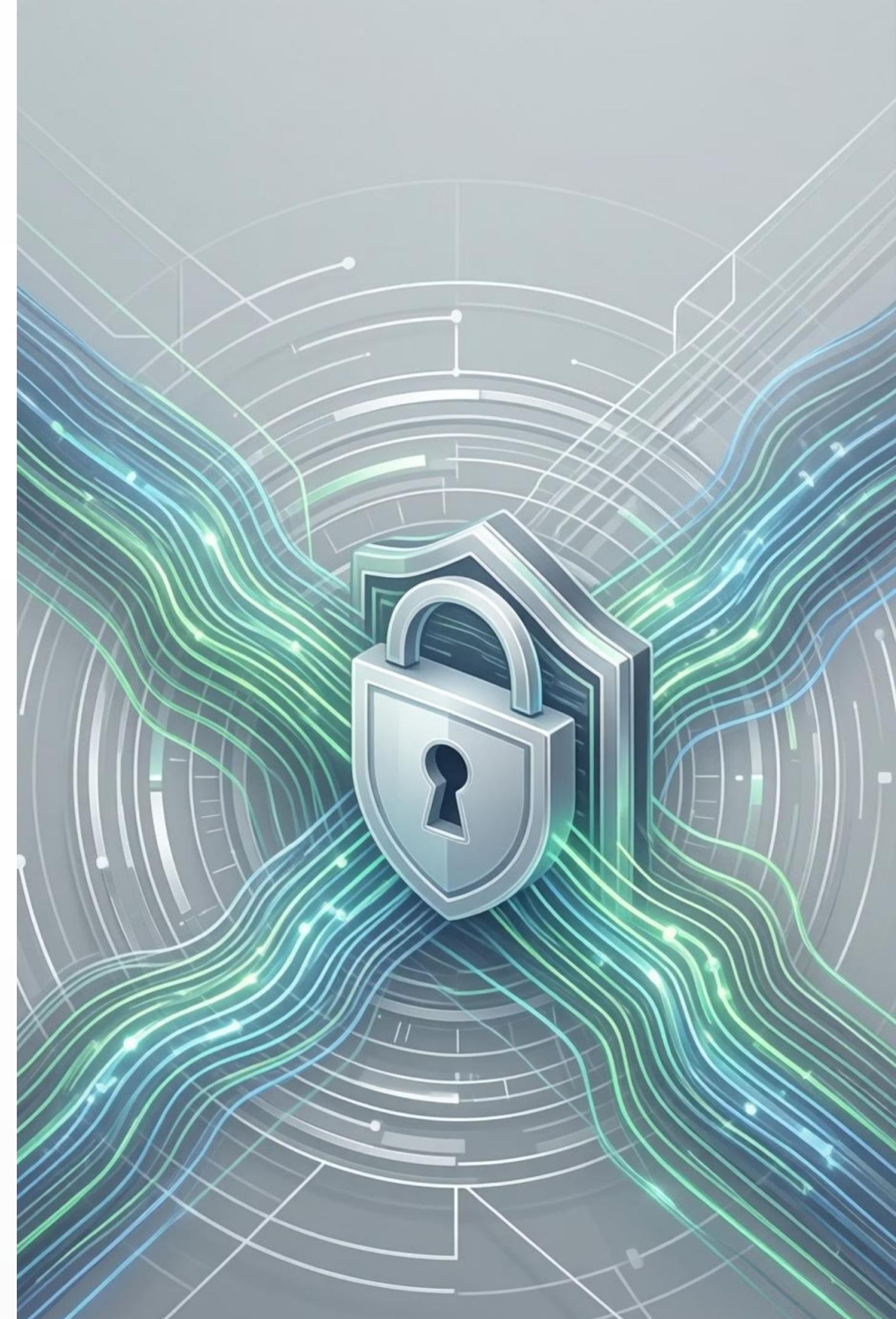
- **Advantages:**

- High security and privacy
- Controlled access to sensitive information
- Improved internal communication
- Faster data transfer (local network)
- Cost-effective for organizations
- Better employee collaboration.



Layered System: Internet, Intranet, and Extranet (Intranet)

- **Disadvantages:**
 - Limited to organization members
 - Requires IT infrastructure and maintenance
 - Initial setup and ongoing costs
 - Less information available compared to Internet





Layered System: Internet, Intranet, and Extranet (Extranet)

An Extranet is an extended intranet that allows controlled access to external partners, suppliers, and clients while maintaining security

- **Key Characteristics:**

- **Scope:** Organization + selected external partners
- **Access:** Restricted - only authorized external users
- **Users:** Employees + business partners/suppliers/clients
- **Security:** High security with controlled access and authentication
- **Ownership:** Owned and maintained by the organization

- **Common Uses:**

- Supplier portals for ordering and inventory
- Client collaboration platforms
- B2B (Business-to-Business) transactions
- Project collaboration with external teams
- Customer support and service portals
- Vendor management systems
- Shared document repositories
- Real-time communication with partners



Layered System: Internet, Intranet, and Extranet (Extranet)

- **Advantages:**

- Secure collaboration with external parties
- Controlled access to sensitive business information
- Improved efficiency in B2B operations
- Reduces need for physical meetings
- Strengthens business relationships
- Cost-effective for multi-organization projects

- **Disadvantages:**

- Complex setup and management
- Requires robust security measures
- Higher maintenance costs
- Potential security risks if not properly managed
- Dependency on partner compliance



Layered System: Internet, Intranet, and Extranet (Internet)

- An Internet refers to a group of networks that are connected to each other.
- Internet also refers to the products, concepts, technologies used to develop the connections among disparate networks.
- **Key Characteristics:**
 - **Scope:** Global - spans across all continents
 - **Access:** Public - anyone with a connection can access
 - **Users:** Billions of people worldwide
 - **Security:** Lower security - open to everyone
 - **Ownership:** No single owner - decentralized
- **Common Uses:**
 - Web browsing (Google, YouTube, Wikipedia)
 - Email communication
 - Social media (Facebook, Twitter, Instagram)
 - Online shopping and banking
 - Streaming services (Netflix, Spotify)
 - Cloud storage and services



Layered System: Internet, Intranet, and Extranet (Internet)

- **Advantages:**
 - Universal access and connectivity
 - Vast amount of information available
 - Enables global communication
 - Supports innovation and collaboration
- **Challenges:**
 - Security and privacy concerns
 - Cybersecurity threats
 - Digital divide (unequal access)
 - Information overload

Internet Access Technologies: Wired Connections

Internet access technologies are methods that enable users to connect to the Internet. Each technology offers different speeds, costs, and availability.



Dial-up

Speed: 56 Kbps
Uses: Telephone lines
Status: Obsolete
Brief: Connects through phone line, slow, ties up phone



DSL (Digital Subscriber Line)

Speed: 1-100 Mbps
Uses: Telephone lines with special equipment
Best for: Basic browsing and email
Brief: Faster than dial-up, doesn't block phone line, distance affects speed



Cable Internet

Speed: 10-500 Mbps
Uses: Cable TV infrastructure
Best for: Streaming and gaming
Brief: Fast and reliable, shared bandwidth with neighbors



Fiber Optic

Speed: 100-1000+ Mbps
Uses: Fiber optic cables
Best for: Heavy users and businesses
Brief: Fastest option, uses light pulses, limited availability

Internet Access Technologies: Wireless & Satellite



WiFi

Speed: 11-1000+ Mbps

Range: 30-100 meters

Best for: Home and office networks

Brief: Wireless, convenient, affected by obstacles and interference



Mobile Networks (4G/5G)

Speed: 4G: 10-100 Mbps, 5G: 100-1000+ Mbps

Coverage: Nationwide

Best for: Mobile devices and on-the-go

Brief: Cellular networks, always connected, requires subscription



Satellite

Speed: 25-100 Mbps

Coverage: Global

Best for: Remote areas

Brief: Global coverage, high latency, affected by weather





World Wide Web (WWW)

- The world wide web (WWW) is defined as a distributed client-server service , in which a client using a browser can access a service using a server .
- However, the service provided is distributed over many locations called sites.
- Each site hold one or more documents, referred as web pages. These documents are connected together through the use of hyperlinks or hypertext.
- Hyperlink is a text which contains links to other texts and are created with hypertext markup language such as HTML.

The Internet

- **Definition:** Global network of interconnected computers
- **Scope:** Infrastructure and physical network
- **Purpose:** Enables data transmission Between devices
- **Technologies:** TCP/IP protocols, routers, servers
- **Services:** Email, FTP, VoIP, Web, etc.

The Web (WWW)

- **Definition:** System of interconnected documents and resources
- **Scope:** Application layer on top of Internet
- **Purpose:** Share and access information through hyperlinks
- **Technologies:** HTTP/HTTPS, HTML, browsers
- **Services:** Web pages, websites, web applications

World Wide Web (WWW)

Web Generations: Evolution of the World Wide Web

Web 1.0 (1991-2004)

The Static Web

- Read-only content with static HTML pages
- Limited user interaction; one-way communication
- Examples: Early informational websites, online brochures
- Characteristics: Simple interface, text-heavy, slow connections.
- Key areas of interest centered on protocols such as HTTP, open standard markup languages such as HTML and XML.

Web 2.0 (2004-Present)

The Social Web

- User-generated content and interactive, dynamic pages
- Enabled social media platforms and two-way communication
- Examples: Facebook, YouTube, Twitter, Wikipedia
- Characteristics: Collaborative, real-time updates, mobile-friendly

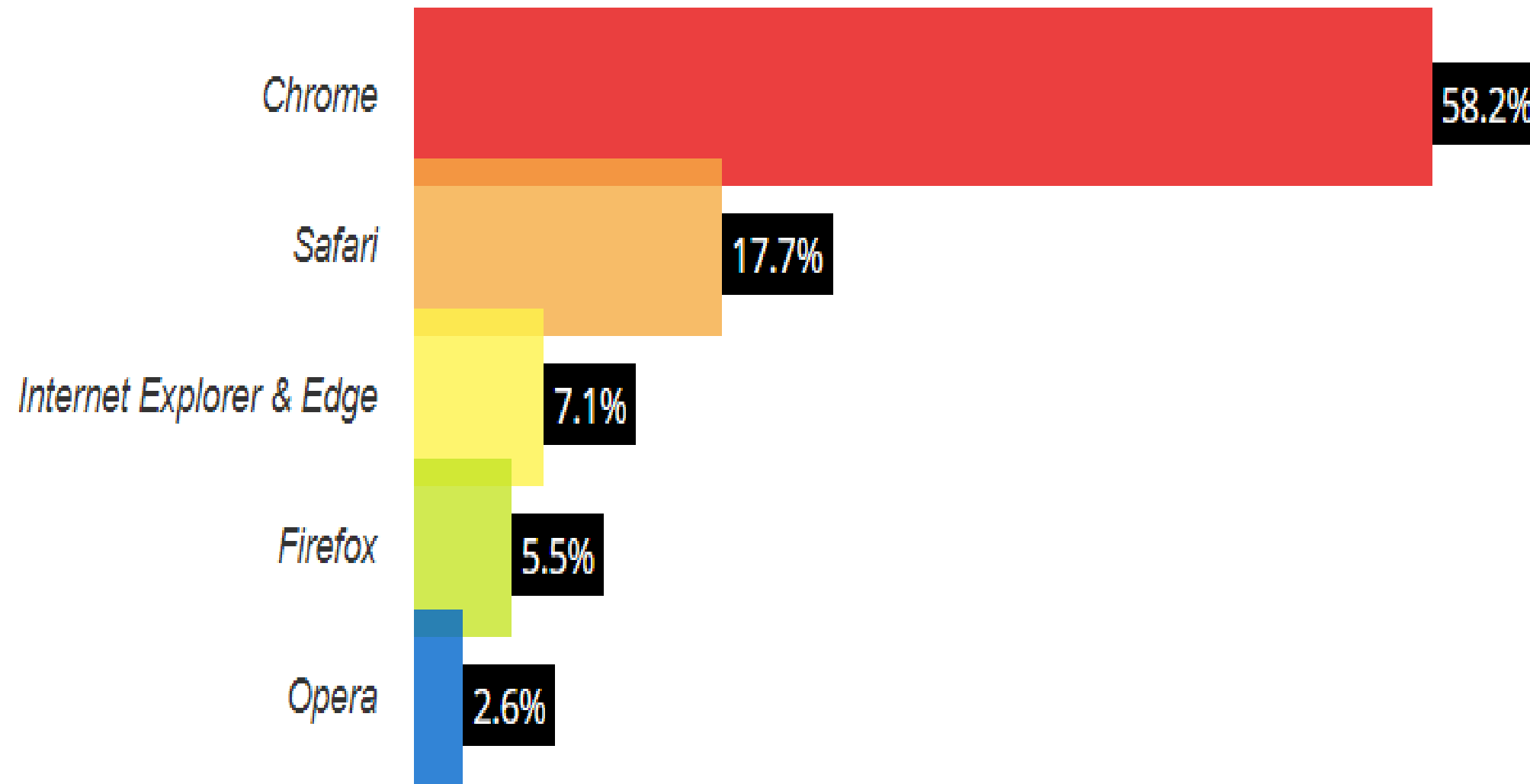
Web 3.0 (2006-Present)

The Semantic Web

- Leverages AI and Machine Learning for intelligent interactions
- Built on decentralized networks and blockchain technology
- Examples: AI assistants, personalized recommendations, metaverse experiences
- Characteristics: Intelligent, autonomous, user-centric, interconnected data

World Wide Web (WWW)

Popular Web Browsers



World Wide Web (WWW)

How the Web Works

1. The web information is stored in the web pages in HTML format.
2. The web pages are stored in web server file system.
3. The web client with specific web browser reads the pages.
4. The web server waits for the request from the web client over the internet (IIS , Apache).

Main Components:

- URLs - Web Addresses
- Client-Side - Browser and User
- Server-Side - Web Servers and Processing
- Request-Response Cycle
- HTTP/HTTPS Protocols



World Wide Web (WWW)

URLs: Web Addresses

A Uniform Resource Locator is the unique address that tells your browser where to find any resource on the web.

URL Structure Breakdown:

Protocol://Domain/Path?Query#Fragment

- **Protocol:** `https://` (data transfer method)
- **Domain:** `www.example.com` (website's name)
- **Path:** `/pages/about` (specific resource location)
- **Query:** `?id=123` (data sent to server)
- **Fragment:** `#section1` (specific part of a page)

Example: `https://www.example.com/products?category=books#reviews`

Domains Examples:

- `.com` : commercial
- `.net` : network
- `.edu` : educational
- `.org` : organization
- `.gov` : government

World Wide Web (WWW)

Client-Side: Browser and User

The "client-side" refers to everything that happens within a user's web browser, enabling interactive and dynamic experiences.

Main Components:

- **Web Browser:** Interprets and displays web pages.
- **HTML:** Provides the structure and content.
- **CSS:** Defines the styling and layout.
- **JavaScript:** Adds interactivity and dynamic behavior.

What Happens on Client-Side:

1. User enters a URL.
2. Browser requests and receives HTML, CSS, and JavaScript.
3. Browser renders the page visually.
4. JavaScript executes for interactive features.
5. User interacts directly with the displayed content.

Advantages:

- Fast response to user actions.
- Reduces server processing load.
- Allows for rich, interactive experiences.
- Offline capabilities for some applications.





World Wide Web (WWW)

Server-Side: Web Servers and Processing

The "server-side" is the hidden backend of the web, handling data processing, logic, and content generation before it reaches your browser.

Main Components:

- **Web Server:** Stores and serves web pages.
- **Backend Code:** Processes requests and application logic.
- **Database:** Manages and stores application data.
- **Languages:** PHP, Python, Node.js, Java, etc.

Process Flow:

1. Server receives client request.
2. Processes request, interacts with database.
3. Generates response (e.g., HTML, JSON).
4. Sends response back to client.

Key Advantages:

- Secure data management.
- User authentication & validation.
- Dynamic content generation & scalability.

World Wide Web (WWW)

Request-Response Cycle

This fundamental communication pattern dictates how web browsers (clients) interact with web servers to retrieve and display content.

The Cycle Steps:

1. User initiates action (e.g., URL entry, link click).
2. Browser performs DNS lookup to get IP address.
3. Browser sends HTTP Request to server.
4. Server processes request and prepares data.
5. Server sends HTTP Response back to browser.
6. Browser renders content for user.

Request Contains:

- **Method:** (GET, POST, PUT, DELETE)
- **URL/Path:** Resource location
- **Headers:** Metadata (e.g., content type)
- **Body:** Data (if applicable)

Response Contains:

- **Status Code:** (e.g., 200 Success, 404 Not Found)
- **Headers:** Metadata (e.g., server info)
- **Body:** Content (e.g., HTML, JSON)



World Wide Web (WWW)

Internet Standards and Organizations

The internet's seamless operation relies on global standards. These are developed and maintained by dedicated organizations, ensuring interoperability, security, and open access for everyone.



IETF

Defines fundamental protocols that make the internet work, publishing specifications as Requests for Comments (RFCs).



W3C

Develops open standards for the World Wide Web, such as HTML, CSS, and Accessibility Guidelines.



ICANN

Coordinates the global Internet's unique identifiers, ensuring a stable and secure addressing system like domain names.



THANK
YOU